

Pharma Agentic AI – Analysis Report

Automated opportunity assessment powered by Agentic AI (Master + Worker agents).

Query	Assess the Lung Cancer market opportunity for MOLH1 in Australia with focus on CAGR, competitor density, clinical trials and IP barriers.		
Molecule	MOLH1	Therapy area	Lung Cancer
Region / Country	Australia	Agents run	MARKET EXIM PATENT TRIALS INTERNAL WEB
Generated on	08 Dec 2025, 10:56 PM		

Executive Summary

- Market & growth view: Total market size: 9730.4 Mn USD with average 5Y CAGR 5.5%. Records: 19.
- Supply and trade dynamics: Total imports: 25899.5 tonnes, total exports: 27040.4 tonnes across 253 records.
- IP barriers and freedom-to-operate: Found 151 relevant patents. Earliest expiry year: 2005.
- Clinical development and pipeline: Found 19 clinical trials matching the criteria.
- Internal strategy and unmet needs: MOLH1 Lung Cancer Opportunity in Australia
- Guidelines and real-world evidence: The LUNG 2024 guideline emphasizes step-wise treatment based on disease severity and patient phenotype, with combination therapy recommended for inadequate monotherapy control. Real-world evidence highlights suboptimal adherence, regional variation in outcomes, and unmet needs in symptom control and quality of life, underscoring the importance of addressing access, affordability, and health literacy.

Agent-level Insights

IQVIA Insights Agent

High-level summary

Total market size: 9730.4 Mn USD with average 5Y CAGR 5.5%. Records: 19.

Structured details

Table:

- **molecule:** MOLH1 **indication:** Lung Cancer **country:** Australia **year:** 2023 **sales_usd_mn:** 624.16 **cagr_5y:** 1.65 **competitors:** 11 **patient_burden_index:** 0.59
- **molecule:** MOLH1 **indication:** Lung Cancer **country:** Australia **year:** 2022 **sales_usd_mn:** 822.84 **cagr_5y:** -2.9 **competitors:** 11 **patient_burden_index:** 0.27
- **molecule:** MOLH1 **indication:** Lung Cancer **country:** Australia **year:** 2017 **sales_usd_mn:** 662.26 **cagr_5y:** -0.7 **competitors:** 14 **patient_burden_index:** 0.81
- **molecule:** MOLH1 **indication:** Lung Cancer **country:** Australia **year:** 2016 **sales_usd_mn:** 271.6 **cagr_5y:** 15.69 **competitors:** 4 **patient_burden_index:** 0.48
- **molecule:** MOLH1 **indication:** Lung Cancer **country:** Australia **year:** 2021 **sales_usd_mn:** 106.42 **cagr_5y:** 8.21 **competitors:** 5 **patient_burden_index:** 0.81
- ... (5 more records)

EXIM Trade Agent

High-level summary

Total imports: 25899.5 tonnes, total exports: 27040.4 tonnes across 253 records.

Structured details

Table:

- **molecule:** MOLH1 **formulation:** Ointment **country:** Australia **year:** 2016 **import_tonnes:** 95.99 **export_tonnes:** 40.07
- **molecule:** MOLH1 **formulation:** Syrup **country:** Australia **year:** 2024 **import_tonnes:** 30.48 **export_tonnes:** 144.92
- **molecule:** MOLH1 **formulation:** Inhaler **country:** Australia **year:** 2017 **import_tonnes:** 144.88 **export_tonnes:** 25.58

- **molecule:** MOLH1 **formulation:** Patch **country:** Australia **year:** 2019 **import_tonnes:** 74.76 **export_tonnes:** 42.31
- **molecule:** MOLH1 **formulation:** Injection **country:** Australia **year:** 2016 **import_tonnes:** 89.73 **export_tonnes:** 28.97
- ... (5 more records)

Patent Landscape Agent

High-level summary

Found 151 relevant patents. Earliest expiry year: 2005.

Structured details

Patents:

- **patent_id:** US1000444 **molecule:** MOLH1 **indication:** Lung Cancer **assignee:** PrimeCare **filing_year:** 2013 **expiry_year:** 2025 **status:** Granted **fto_flag:** Low
- **patent_id:** US1000823 **molecule:** MOLH1 **indication:** Lung Cancer **assignee:** NovaCure **filing_year:** 2007 **expiry_year:** 2020 **status:** Pending **fto_flag:** Low
- **patent_id:** US1001818 **molecule:** MOLH1 **indication:** Lung Cancer **assignee:** Sunrise Therapeutics **filing_year:** 1997 **expiry_year:** 2020 **status:** Pending **fto_flag:** Medium
- **patent_id:** US1002894 **molecule:** MOLH1 **indication:** Lung Cancer **assignee:** PrimeCare **filing_year:** 2017 **expiry_year:** 2040 **status:** Pending **fto_flag:** Low
- **patent_id:** US1003274 **molecule:** MOLH1 **indication:** Lung Cancer **assignee:** OncoGenix **filing_year:** 1998 **expiry_year:** 2011 **status:** Expired **fto_flag:** Low
- ... (5 more records)

Clinical Trials Agent

High-level summary

Found 19 clinical trials matching the criteria.

Structured details

Table:

- **trial_id:** CT-007766 **molecule:** MOLH1 **indication:** Lung Cancer **phase:** Phase I **status:** Active **sponsor:** XYZ Pharma **country:** Australia
- **trial_id:** CT-009959 **molecule:** MOLH1 **indication:** Lung Cancer **phase:** Phase III **status:** Terminated **sponsor:** MedLife Corp **country:** Australia
- **trial_id:** CT-012165 **molecule:** MOLH1 **indication:** Lung Cancer **phase:** Phase I **status:** Terminated **sponsor:** ABC Pharma **country:** Australia
- **trial_id:** CT-015426 **molecule:** MOLH1 **indication:** Lung Cancer **phase:** Phase IV **status:** Completed **sponsor:** OncoGenix **country:** Australia
- **trial_id:** CT-019127 **molecule:** MOLH1 **indication:** Lung Cancer **phase:** Phase III **status:** Active **sponsor:** MedLife Corp **country:** Australia
- ... (5 more records)

Internal Knowledge Agent

High-level summary

MOLH1 Lung Cancer Opportunity in Australia

Structured details

Documents:

- asthma_strategy_2024.pdf
- respiratory_field_insights.pdf

Key Opportunities:

- Address unmet need for targeted therapy in Australia's Lung Cancer market
- Leverage MOLH1's efficacy in steroid-intolerant asthma patients for a new indication in Lung Cancer
- Explore fixed-dose combinations with existing asthma therapies for a safer, more adherence-friendly treatment option
- Target pediatric and elderly Lung Cancer patients, where lower-dose treatment options are scarce

- Key Risks:**
- Intense competition in Lung Cancer market with established players and high R&D costs
 - Evidence gaps in MOLH1's clinical efficacy for Lung Cancer indication and potential regulatory hurdles
 - Operational constraints in expanding to Tier-2/3 cities in Australia, where inhaler affordability and accessibility are key issues

Web Intelligence Agent

High-level summary

The LUNG 2024 guideline emphasizes step-wise treatment based on disease severity and patient phenotype, with combination therapy recommended for inadequate monotherapy control. Real-world evidence highlights suboptimal adherence, regional variation in outcomes, and unmet needs in symptom control and quality of life, underscoring the importance of addressing access, affordability, and health literacy.

Structured details

- Links:**
- **title:** Lung Cancer guideline (synthetic) **url:** <https://example-guideline.com>
 - **title:** Lung Cancer RWE summary **url:** <https://example-rwe.com>

- Key Updates:**
- Combination therapy is recommended for lung cancer treatment when monotherapy fails to achieve adequate control.

- Lifestyle, diet, exercise, and periodic monitoring are emphasized as essential components of lung cancer management.

This report is generated by the Pharma Agentic AI prototype for EY Hackathon 6.0. Insights are based on simulated / mock data sources and should not be used as a substitute for validated market research.